Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, May 12, 2022
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. **Walking on a disk**

   A horizontal circular disk of mass M (distributed at uniform density) is free to rotate about a vertical axis which passes through a point P on its rim (attention: NOT through its center). A dog of mass m walks once around the rim (starting from rest). Let $\alpha(t)$ be the angle by which the disk has rotated by time t, and $\beta(t)$ the angle of the dog's position on the rim, relative to the rim.

   a) List the constants of this motion.

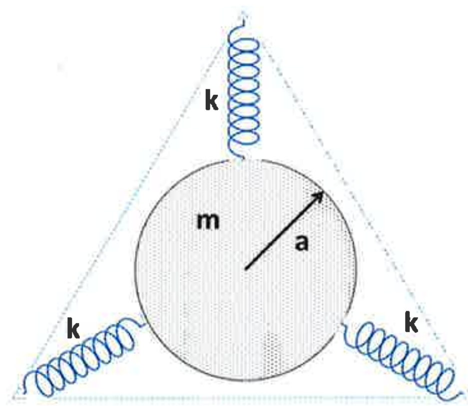
      If you do not recall the computation of the disk's moment of inertia relative to P then just denote it by I_P.

   b) Derive a relation between the rate of change of the angular displacement of the disk (α) to that of the dog relative to the disk. (β).

   c) Compute, in terms of an explicit integral expression, the angle by which the disk turns by the time the dog completes his walk around the rim.

2. **Disk with Three Springs**

A uniform disk of mass m and radius a rests on a horizontal frictionless surface. It is symmetrically attached to three identical, ideal, massless springs of spring constant k whose other ends are attached to the three vertices of an equilateral triangle.



At equilibrium, the length ℓ of the springs is greater than their relaxed length ℓ_0. The disk remains in the initial horizontal plane but is otherwise free to move. (The diagram shows the view looking down on the plane.) What are the frequencies of the normal modes of small oscillations? What do the modes look like?

3. **Sliding rod**

A rigid thin straight rod of length ℓ and mass M, distributed uniformly along its length, is placed leaning on a wall with its top at height h and starts to slide. Ignoring the friction at the contact points with the wall and the floor.

a) Derive the equations by which the rod's motion can be described, and list the constants of motion.

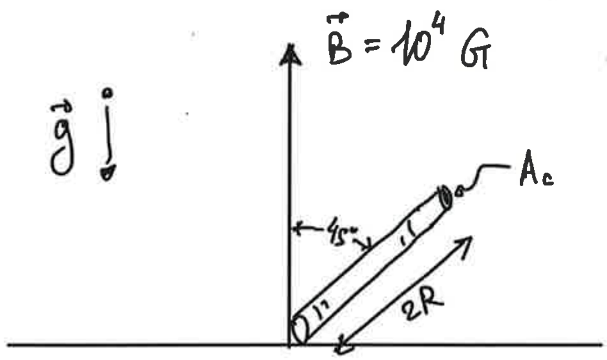
b) At what height would the rod lose contact with the wall?

Section B. Electricity and Magnetism

1. **Ring falling in a magnetic field**

   A round metal ring, is placed on edge at an angle θ_0 of $45°$ to the vertical, on top of a massive magnet, of (the rather high) magnetic field 10^4 Gauss, as shown in the figure. Once released, the ring falls over *without slipping*. The ring is made of silver.

   Silver has a conductivity $\sigma = 6 \times 10^7 \text{sec}^{-1}$, and density $\rho = 10.5 \text{gm/cm}^3$. The ring is of radius R of 1cm and a cross-sectional area A_c, where $A_c^{1/2} \ll R$.

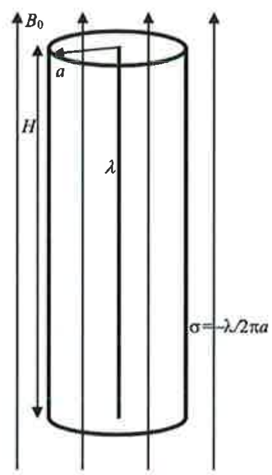


   a) Write down a differential equation for the descent of the ring (i.e. for the angle $\theta(t)$). Show that one can ignore here the inertial term for the values of the parameters given.

   b) Calculate how long it takes the ring to fall over. Note that $\int \frac{d\theta}{\sin(\theta)} = \log(\tan(\theta/2))$.

2. **Spinning Cylinder**

A cylindrical capacitor consists of a line of charge with linear charge density λ and a concentric insulating tube of radius a with a compensating uniform surface charge density $\sigma = -\lambda/2\pi a$ on its surface (fixed, not free to move). The height of the capacitor $H \gg a$ so that you can ignore edge effects. The capacitor is placed in a uniform external magnetic field of strength B_0 parallel to the cylinder axis and pointing up. The insulating tube is free to rotate around its axis and its mass is all concentrated on the rim.



a) Find the magnitude and direction of the electromagnetic angular momentum stored in the EM field.

b) The external magnetic field B_0 is very slowly ramped down. Show that this will cause the tube to rotate and find the angular velocity of rotation when the external B field is completely turned off.

3. **A conducting sphere in an electric field**

An insulated, uncharged, conducting spherical shell of radius a is placed in a uniform electric field of magnitude E_0.

a) Show that the electrostatic potential outside the sphere due to the induced charges equals that produced by a dipole placed at its center.

Hint: it is useful (though not absolutely necessary) to recall that a charge Q placed at distance R from the center of a grounded sphere induces what looks as a mirror charge $-Qa/R$ at a point at distance a^2/R from the center.

b) Suppose the shell is cut into two hemispheres at its equator (in the plane perpendicular to the field). What force is required to keep the hemispheres from separating?

(Explain your calculation.)

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Graduate Preliminary Examination
Part II

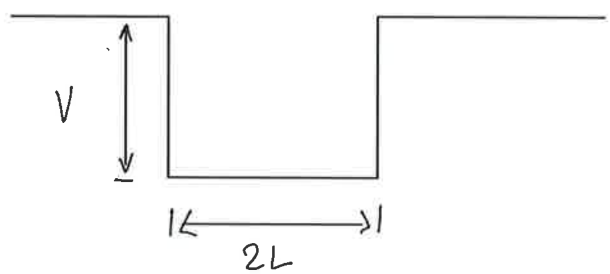
Friday, May 13, 2022
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. A particle of mass m is confined to a one-dimensional square potential well of depth V and width $2L$. The parameters V and L are such that there is only one bound state in this well. Assume that $\hbar/L\sqrt{mV}$ is very large. You may wish to sketch the wavefunction in this limit. If the particle is initially in this state, and the width of the well is suddenly decreased to L symmetrically, calculate the probability that the particle will escape from the well.



2. **Tritium Beta Decay**

Tritium is radioactive and decays to once-ionized He^3 with the emission of an electron ($T \rightarrow^3 He^+ + e^- + \nu$). You can assume that, as far as the atomic electron is concerned, all that happens is that the nucleus suddenly changes its charge from +1 to +2. Just before the decay the atomic electron is in the hydrogen ground state wave function $\psi_0 = \psi^{Z=1}_{n=1,\ell=0}$ and it remains in that state immediately after the decay.

(a) Assuming that the tritium atom is originally in its ground state, what is the probability of finding the resulting He^+ ion in its $1s$ ground state immediately after the decay?

(b) Show that the probability of finding the He^+ ion in any of its three $2p$ excited states immediately after the decay is zero. Find the non-zero probability of being in the $2s$ state.

(c) Some questions about energy (which can be answered without doing any integrals): What is the expectation value of the atomic electron's energy ϵ_{at} immediately after the decay? If, after the decay, the atomic electron is in an excited state, it will soon decay back to the ground state with a photon carrying away the difference in energy with the ground state. Assuming that the atomic electron has negligible probability of being in an unbound state after the beta decay (not too far from true), what is the average energy carried off by the post-decay photon?

You will need some hydrogen-like wave functions:

$$\phi_{1s} = \frac{1}{\sqrt{\pi a_B^3}} e^{-r/a_B}, \phi_{2s} \propto (1 - r/2a_B)e^{-r/2a_B}, \ \phi_{2p,0} \propto z e^{-r/2a_B}, \ \phi_{2p,\pm 1} \propto (x \pm iy) e^{-r/2a_B}$$

where a_B is the Bohr radius for a hydrogen-like atom of central charge Z: $a_B = a_0/Z$.

3. The following is a model for electronic states in ring-shaped molecules, such as benzene. Two identical fermions of mass M and spin $1/2$ are confined to a circular ring of radius R.

   The Hamiltonian for the two-particle system is

$$H = -\frac{\hbar^2}{2MR^2}\sum_{j=1,2}\frac{\partial}{\partial\theta_j^2} + A\cos(\theta_1 - \theta_2) \equiv H_0 + V$$

   where θ_j is the position of the particle jth measured as an angle around the circle. Assume that A is much less than $\frac{\hbar^2}{2MR^2}$.

   Find the energies, degeneracies, and wavefunctions, to the first order in $\lambda = A/\frac{\hbar^2}{2MR^2}$, of:

   (a) The ground state(s)

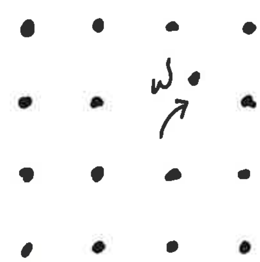
   (b) The first excited state(s), i.e. states of the first energy above that of (a).

   Remarks: remember to include in your considerations also the spin degrees of freedom, and state clearly the relevant considerations.

Section B. Statistical Mechanics and Thermodynamics

1.

A crystal consists of N ions placed on N lattice sites. The ions can absorb a phonon of frequencies ω and be displaced to any one of the N' interstitial sites ($N' > N$) as shown in the figure below.



a) Calculate the entropy for the state containing n defects, where $1 \ll n \ll N, N'$.

b) If the system at equilibrium at a temperature T, find the number of defects n. Express your answer in terms of N, N', ω, T and fundamental constants.

2. **Atmospheric density**

   (a) If you hold up your hand, how many molecules per second are colliding with your palm? We are interested in the order of magnitude, not the last factor of two. Approximate the atmosphere as pure oxygen (of molecular mass 32 a.m.u.).

Suppose that the atmosphere was in equilibrium at constant temperature. Then, in addition to the kinetic energy of the molecules in the air, one would have to take account of the potential energy due to gravity. Under these assumptions:

   (b) What is the probability distribution for the height h of a molecule above the surface of the earth?

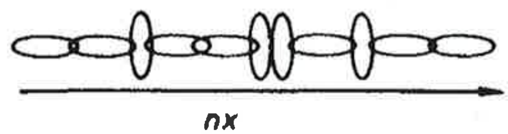
   (c) How much less oxygen would you find at a height of 1 km than you do at the earth's surface?

Some relevant constants: $R \approx 8.3\ \frac{\mathrm{J}}{\mathrm{mol\ K}}$, $N_A \approx 6.02 \times 10^{23} \mathrm{mol}^{-1}$.

3. **An elastic chain**

Consider a one-dimensional chain consisting of $n \gg 1$ segments (as in the figure). Each link can be in one of two states: i) aligned with the chain, in which case its contribution to the chain's extension is a, or ii) transversal to the chain, in which its contribution to the chain's extension is 0.

Denote by X be the the chain's extension ($= a\times$ number of "up" segments). The work spend to stretch the chain to length X by applying to it force F, is $W = X \times F$.



a) Calculate the chain's free energy $\mathcal{F}(T, F) = FX - T\mathcal{S}$ at temperature T, when it is under tension F. ($\mathcal{S}$ denotes here the system's entropy.)
Hint: one may start by calculating the partition function under the applied tension.

b) Calculate the thermodynamic relation between the tension F and the elongation X, when the chain is maintained at temperature T.

c) Under what (limiting) conditions does the chain display Hook's law, $X = (-)\lambda(T)F$?

d) Does this spring constant $\lambda(T)$ increase or decrease with the temperature?